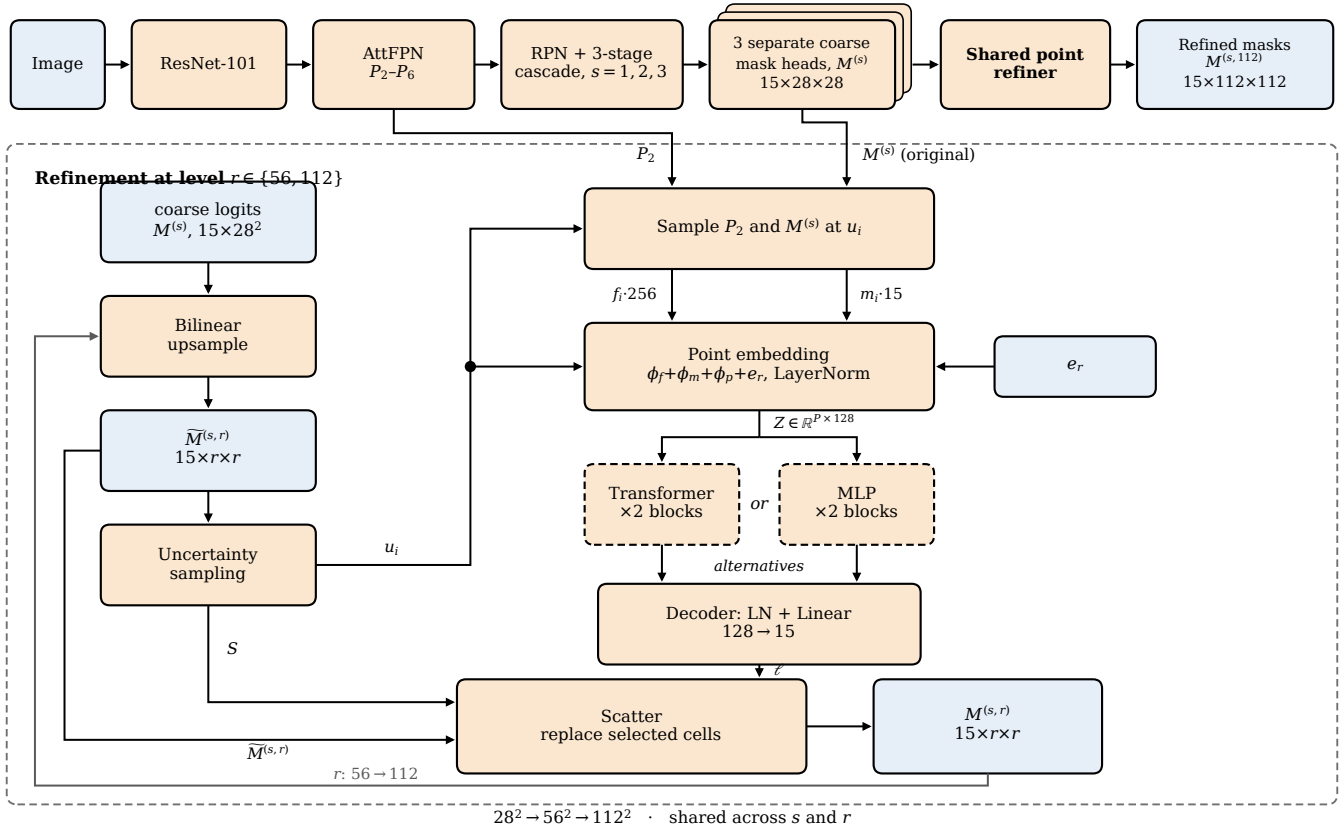
Image
ResNet-101
AttFPN
P_2–P_6
RPN + 3-stage cascade, $s = 1, 2, 3$
3 separate coarse mask heads, $M^{(s)}$
15×28×28
Shared point refiner
Refined masks
$M^{(s,\,112)}$
15×112×112
P_2
$M^{(s)}$ (original)
Refinement at level $r \in \{56, 112\}$
coarse logits
$M^{(s)}$, 15×28^2
Bilinear upsample
$\widetilde{M}^{(s,\,r)}$
$15\times r\times r$
Uncertainty sampling
u_i
S
Sample P_2 and $M^{(s)}$ at u_i
$f_i \cdot 256$
$m_i \cdot 15$
Point embedding
$\phi_f+\phi_m+\phi_p+e_r$, LayerNorm
e_r
$Z \in \mathbb{R}^{P\times128}$
Transformer ×2 blocks
or
MLP ×2 blocks
alternatives
Decoder: LN + Linear
$128 \to 15$
ℓ
Scatter
replace selected cells
$M^{(s,\,r)}$
$15\times r\times r$
$\widetilde{M}^{(s,\,r)}$
r: $56 \to 112$
$28^2 \to 56^2 \to 112^2$ · shared across s and r